# CDAX Student Complaint Root Cause Analysis

A structured analytical model to classify complaints, identify recurring issues, and prioritize corrective actions

| COMPLAINT RECORDS | CONFIRMED ROOT CAUSES | AVAILABLE EVIDENCE |
|---|---|---|
| **Not verified** | **Not calculable yet** | **Support tracking specified** |

**Important data note:** Available CDaX documentation describes support requests as an event-tracking requirement, but actual complaint/ticket records were not verified in the reviewed sample. Therefore, this report does not invent complaint counts, course rankings, severity percentages, resolution rates, or root causes. It provides the analysis framework and investigation plan.

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# 1. Objective and Scope

Develop a repeatable method for understanding student complaints across CDaX, finding recurring issues, separating symptoms from confirmed causes, and assigning measurable corrective actions. The reviewed sample and documentation do not support actual complaint-frequency or resolution-performance calculations. Missing sample records do not prove that a complaint feature is absent from the live app.

# 2. Complaint Categories and Root-Cause Hypotheses

| Category | Examples | Potential factors to investigate (not confirmed causes) | Owner |
|---|---|---|---|
| Academic / learning | Difficult topic, unclear lesson, quiz/project guidance | Prerequisite gaps, content sequence, question ambiguity, mentor availability | Academic / Content |
| Technical / app | Login, page error, video playback, progress not saved | API errors, device compatibility, session handling, sync or video delivery | Engineering / QA |
| Administrative | Enrollment status, schedule, certificate or account request | Manual processing, unclear policy, delayed status updates | Operations |
| Payment / subscription | Failed or duplicate payment, purchase not reflected, refund | Gateway response, callback handling, reconciliation, refund workflow | Finance / Engineering |
| Live class / demo | Join link, schedule, demo booking, attendance mismatch | Scheduling, notification delivery, attendance event capture | Operations / Engineering |
| Mentor / support | Slow response, unresolved query, repeated follow-up | Support capacity, ticket routing, ownership, escalation process | Support / Academic |
| Other / unclassified | Issue outside categories or insufficient details | Taxonomy gap or incomplete complaint description | Support triage |

Use one primary category per complaint. Optional tags can capture secondary factors. Preserve the original complaint text so classification decisions can be reviewed.

# 3. Required Complaint Data

| Field | Purpose |
|---|---|
| complaint_id | Unique ID; required for deduplication and tracking |
| student_id | Pseudonymous key to identify repeat contacts; restrict access |
| created_at | Timestamp and consistent timezone |
| course_id / module_id | Course/module involved, when applicable |
| category / subcategory | Academic, technical, administrative, payment, live class, support, other |
| description / issue_code | Original text plus standardized issue label |
| severity / status | Critical, High, Medium, Low; Open, In Progress, Waiting, Resolved, Closed |
| assigned_team / owner | Accountable team or person |
| resolved_at / resolution_code | Resolution timestamp and standardized outcome |
| root_cause_code | Confirmed cause only; otherwise mark pending investigation |
| duplicate_of / repeat_group_id | Links repeat reports about the same underlying issue |
| student_impact | Learning blocked, payment at risk, inconvenience, etc. |
| corrective_action / due_date | Action, owner, deadline, verification result |
| source_channel | In-app, email, chat, phone, class, or other verified channel |

# 4. Analytical Measures and Formulas

Calculate only after actual complaint records are supplied and validated. Missing values must remain blank or be reported as unavailable, not treated as zero.

| Measure | Calculation | Use |
|---|---|---|
| Total complaints | Count distinct complaint_id | Overall recorded volume for the reporting period |
| Category share | Category count ÷ valid complaint count × 100 | Shows concentration by type |
| Course/module frequency | Count complaints grouped by course_id/module_id | Highlights areas with higher raw counts |
| Complaint rate | Course/module complaints ÷ enrolled/active students × 100 | Allows comparison across different course sizes; requires reliable denominator |
| High-severity share | High + Critical ÷ complaints with known severity × 100 | Tracks urgent student-impact issues |
| Open backlog | Count where status is not Resolved/Closed | Unresolved workload at reporting date |
| Resolution time | resolved_at – created_at; report median and 90th percentile | Shows how long resolution takes |
| SLA compliance | Resolved within approved SLA ÷ complaints due × 100 | Measures response/resolution target performance |
| Repeat issue rate | Complaints in repeat groups ÷ valid complaints × 100 | Finds recurring issues |
| Reopen rate | Reopened resolved complaints ÷ resolved complaints × 100 | Signals ineffective fixes |
| Root-cause concentration | Complaints by confirmed cause ÷ complaints with confirmed cause | Prioritizes systemic causes |

### Recommended breakdowns

Review by week/month, course, module, category, severity, status, owner, source channel, confirmed root cause, and repeat group. Normalize counts by enrolled/active students, class sessions, or payment transactions where appropriate. A higher raw count may simply reflect a larger student population.

# 5. Severity and Prioritization

| Level | Suggested definition | Suggested response target* | Escalation |
|---|---|---|---|
| Critical | Broad outage, privacy/safety concern, material financial harm, or many students blocked | Immediate triage; same-day action plan | Management + relevant owner |
| High | Paid course access blocked, serious payment issue, repeated core workflow failure | Prioritize within 1 business day | Team lead |
| Medium | Important function degraded but workaround exists; progress affected | Review within 3 business days | Assigned owner |
| Low | Minor inconvenience or information request without material impact | Normal queue under agreed SLA | Standard queue |

*Proposed starting targets for approval, not existing CDaX service commitments.

### Suggested priority score

Priority Score = Severity (1–4) × Frequency (1–5) × Repeat Pattern (1–3) × Student Impact (1–3). Define the scales before use and review high-scoring issues with owners. This is a proposed method, not a score calculated from current complaints.

# 6. Root-Cause Investigation Method

### Capture

Create a unique complaint ID and record time, course/module, category, severity, status, and original description.

### Normalize

Map similar reports to a stable issue code while preserving original wording.

### Detect recurrence

Group by issue code, module, time window, affected workflow, and repeat_group_id.

### Investigate evidence

Review relevant app logs, API responses, database records, payment reconciliation, class schedules, content, and support history.

### Separate symptom from cause

Example: “progress not saved” is a symptom. Check logs before deciding whether the cause is API failure, session expiry, or client-side sync.

### Assign action

Record corrective action, accountable owner, due date, expected outcome, and verification method.

### Validate after fix

Monitor the same issue and compare post-fix complaint rate and recurrence with the baseline.

# 7. Potential Areas to Investigate

The following are hypotheses based on documented CDaX feature/data areas; none is a confirmed cause of actual complaints.

| Area | Evidence to inspect | Action if confirmed | Success measure |
|---|---|---|---|
| Content / assessment | Lesson feedback, quiz attempts, question wording and content version | Clarify lesson, add prerequisites, review ambiguous questions | Repeat academic complaints; completion/assessment outcomes |
| Login / progress | Frontend errors, API logs, auth/session events, progress-write failures | Fix reproducible errors; improve error messages and retry/sync | Issue rate per active student; progress-sync complaints |
| Payment / enrollment | Gateway responses, purchase records, callbacks and reconciliation | Improve payment-state reconciliation and status messaging | Complaints per transaction; unresolved mismatches |
| Live class / demo | Schedules, join links, attendance events, notification logs | Validate links/timezones, reminders, attendance capture | Complaints per class session |
| Support process | Ticket timestamps, owner changes, repeat contacts and resolution codes | Introduce ownership, escalation, response targets and templates | Backlog, resolution time, reopen/repeat rate |
| Certificate / admin | Enrollment status, completion criteria, certificate records | Clarify eligibility and request status; automate checks if feasible | Processing delays and status-query complaints |

# 8. Recommended Management Dashboard

| Section | Recommended contents |
|---|---|
| Summary cards | Total complaints, open backlog, high/critical complaints, median resolution time, repeat issue rate. Show “data unavailable” where source data is missing. |
| Category and severity | Bar charts of actual complaint counts by category and severity. |
| Course/module | Counts and rates per enrolled/active student where valid denominators exist. |
| Trend and recurrence | Weekly/monthly trend, top repeat groups, reopened complaints, age of open tickets. |
| Root cause and actions | Confirmed cause, evidence/confidence, priority, owner, due date, status, post-fix measure. |

## 9. Corrective-Action Register Template

Use one row per confirmed root cause/action and review it regularly with the relevant product, academic, support, operations, or engineering owner.

| Issue / repeat group | Evidence & confirmed cause | Priority | Corrective action | Owner | Due date | Status | Post-fix KPI |
|---|---|---|---|---|---|---|---|
| Populate from verified records | Populate from verified records | Populate from verified records | Populate from verified records | Populate from verified records | Populate from verified records | Populate from verified records | Populate from verified records |
| Populate from verified records | Populate from verified records | Populate from verified records | Populate from verified records | Populate from verified records | Populate from verified records | Populate from verified records | Populate from verified records |
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## 10. Data Quality Checks

• Every complaint has a unique complaint_id; duplicate submissions are linked rather than silently deleted.

• Category, severity, status, and timestamps use consistent formats and approved values.

• Course/module IDs match the official catalogue where applicable.

• Resolved/closed complaints have a resolution timestamp and resolution code; reopenings are recorded.

• Root cause remains pending until supported by evidence.

• Student identifiers and complaint text are access-controlled; management reporting is aggregated where possible.

• Ticket counts reconcile with the support source; missing events and invalid fields are monitored.

• Reporting period, timezone, inclusion rules, and student/course denominator are documented.

## 11. Conclusions and Next Steps

The available evidence supports defining a structured complaint-analysis model, but not quantifying actual complaint causes or identifying which course has the most complaints. Obtain the real support/complaint export or confirm the production support table/API before reporting measured results.

| Step | Action |
|---|---|
| 1. Confirm source | Request actual ticket/support records from the backend or support owner. |
| 2. Validate fields | Map available fields to this model; document missing fields without fabricating values. |
| 3. Build baseline | Calculate category, course/module, severity, backlog, resolution-time, and recurrence measures. |
| 4. Investigate top issues | Verify the highest-impact and repeated causes using logs and business records. |
| 5. Track improvements | Assign owners and deadlines, then compare post-fix metrics against baseline. |

**Management takeaway:** Use this report as the structured model and investigation plan. Categories, possible causes, response targets, and priority scores are proposed guidance—not measured CDaX results until validated against actual complaint records.

Evidence reviewed: CDaX Data Dictionary and Support entity concept, CDAX Event Tracking Specification, and available sample-data/analytics audit documentation. The reviewed sample did not verify actual support-ticket records. This report is not a production support-system audit.